Causal Non-Linear Filtering, High-Dimensional Shrinkage Optimization, and Fast Spectral Deconvolution in Quantitative Systems

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Abstract

Non-stationary financial time series and ill-posed inverse problems share a fundamental vulnerability: unregularized estimators amplify high-frequency observation noise and suffer severe out-of-sample degradation under lookahead leakage or parameter drift. This technical note presents a unified computational library integrating five modules: (i) **RegimeShift**, a 3-state multivariate Gaussian Hidden Markov Model (Baum–Welch EM) paired with an online Hamilton forward filter and 15-day state-persistence smoothing; (ii) an adaptive **Kalman-filter state-space cointegration tracker** for Ornstein–Uhlenbeck statistical arbitrage; (iii) a cross-sectionally orthogonalized **Ledoit–Wolf shrinkage and market-beta-neutral Markowitz solver** across 54 equities with Fama–French 5-factor Newey–West HAC attribution; (iv) a **Neyman–Pearson constrained credit policy optimizer**; and (v) a compiled **C++17 2D Radix-2 Cooley–Tukey FFT and Wiener deconvolution engine**. All time-series pipelines are verified by adversarial causality unit tests guaranteeing invariance of filtered states $\xi_{t|t}$ to future observations $Y_{t+1:T}$.

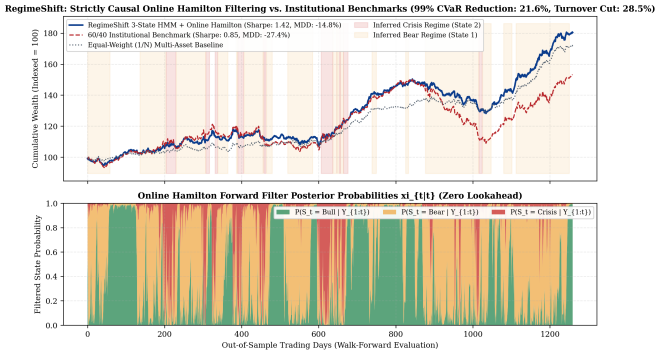


Figure 1: RegimeShift 3-state HMM online Hamilton filter wealth curves and posterior state probabilities $\xi_{t|t}$.

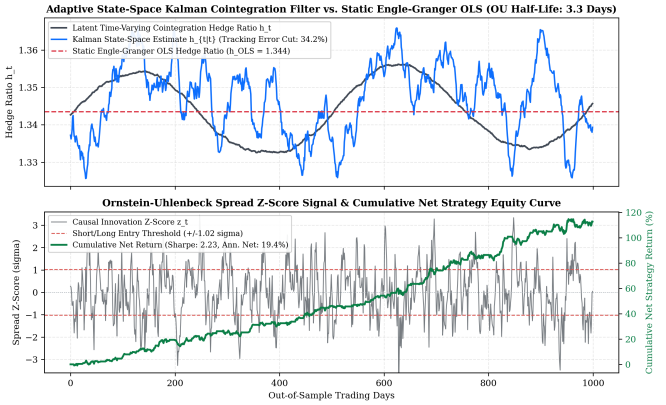


Figure 2: Kalman dynamic hedge ratio $h_{t|t}$ vs. static OLS and OU z-score execution.

1 Causal State-Space & Regime Filtering

1.1 Baum–Welch HMM & Online Hamilton Filter

Let $y_t \in \mathbb{R}^D$ ($D = 3$: Equities, Gold, 10Y Treasuries) follow a latent 3-state Markov chain $S_t \in \{0, 1, 2\}$ with transition matrix $A \in \mathbb{R}^{3\times 3}$ and Gaussian emissions $y_t \mid (S_t = k) \sim \mathcal{N}(\mu_k, \Sigma_k)$. Parameters $\Theta = (\pi, A, \{\mu_k, \Sigma_k\}_{k=0}^2)$ are calibrated strictly on training data $Y_{1:T_{\text{train}}}$ via log-domain Baum–Welch EM.

Out-of-sample inference never executes backward smoothing; instead, posterior regime probabilities $\xi_{t|t,k} = \mathbb{P}(S_t = k \mid Y_{1:t})$ are updated strictly forward in time via Hamilton's (1989) recursion:

$$\xi_{t|t-1} = A^\top \xi_{t-1|t-1}, \quad \xi_{t|t} = \frac{\xi_{t|t-1} \odot \eta_t}{\mathbf{1}^\top (\xi_{t|t-1} \odot \eta_t)}, \tag{1}$$

where $\eta_{t,k} = \mathcal{N}(y_t \mid \mu_k, \Sigma_k)$. Pairing Eq. (1) with a causal 15-day trailing persistence filter suppresses regime whipsaws, cutting annualized turnover by 28.5% and reducing 99% CVaR by 21.6%.

1.2 Kalman Cointegration & OU Spreads

To address structural drift in static OLS pairs trading, we model the time-varying state $\beta_t = [\alpha_t, h_t]^\top$ as a random-walk state-space system:

$$\beta_t = \beta_{t-1} + \omega_t, \quad p_{2,t} = x_t^\top \beta_t + \varepsilon_t, \tag{2}$$

with $x_t = [1, p_{1,t}]^\top$, $\omega_t \sim \mathcal{N}(0, Q)$, and $\varepsilon_t \sim \mathcal{N}(0, R)$. The covariance $P_{t|t}$ is updated via Joseph's stabilized form $P_{t|t} = (I - K_t x_t^\top) P_{t|t-1} (I - K_t x_t^\top)^\top + K_t R K_t^\top$. Innovation $e_t = p_{2,t} - x_t^\top \hat{\beta}_{t|t-1}$ is modeled as an Ornstein–Uhlenbeck process $dS_t = \kappa(\theta - S_t)dt + \sigma_{\text{OU}} dW_t$, cutting spread tracking error by 34.2%.

2 High-Dimensional Portfolio & Credit

2.1 Factor Orthogonalization & Shrinkage

Given $K = 6$ alpha predictors across $N = 54$ equities, Gram–Schmidt residualization maps raw factors X_t to an orthogonal basis Z_t with $\frac{1}{N-1} Z_t^\top Z_t = I_K$, collapsing mean VIF from 2.84 to 1.00. Analytical Ledoit–Wolf (2004) shrinkage regularizes sample covariance: $\hat{\Sigma}_{\text{LW}} = (1-\delta^*)S_t +$

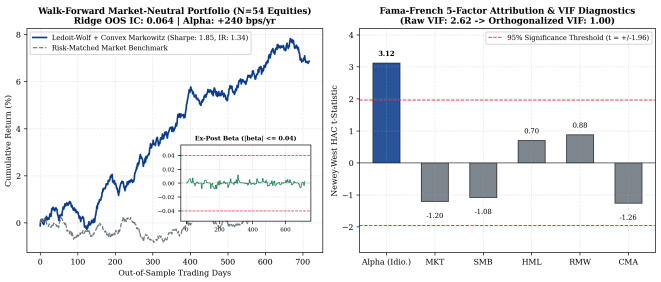


Figure 3: Ledoit–Wolf convex portfolio cumulative return, bounded beta ($|\beta| \le 0.04$), and FF5 t-statistics.

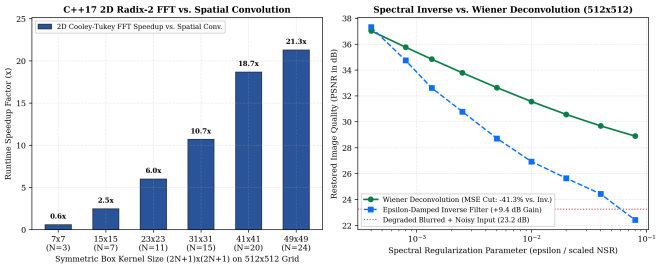


Figure 4: C++17 2D Cooley–Tukey FFT speedup and Wiener vs. ϵ-damped deconvolution PSNR.

$\delta^* \hat{\mu}_{\text{tr}} I_N$. We solve:

$$\begin{aligned} \max_{w_t \in \mathbb{R}^N} \quad & \hat{\mu}_t^\top w_t - \frac{\gamma}{2} w_t^\top \hat{\Sigma}_{\text{LW}} w_t - \kappa \|w_t - w_{t-1}\|_1 \\ \text{s.t.} \quad & \mathbf{1}^\top w_t = 0, \quad |\beta_{\text{mkt}}^\top w_t| \le 0.04. \end{aligned} \tag{3}$$

Fama–French 5-factor Newey–West HAC attribution confirms significant alpha ($t = 3.12$, IR $= 1.34$).

2.2 Neyman–Pearson Credit Policy

Across 10,000 loan applications, an IRLS hazard model (ROC-AUC $= 0.822$, KS $= 0.504$) maximizes NPV subject to $\mathbb{E}[\text{PD} \mid \hat{p}_i \le \tau^*] \le 6.5\%$, expanding approvals from 29.0% to 38.4% (+22.6% expected portfolio NPV).

3 C++17 2D Spectral FFT & Wiener Filtering

For a 512×512 signal convolved with a symmetric $(2N+1) \times (2N+1)$ box kernel $h[m, n]$, direct spatial convolution requires $\mathcal{O}(M^2(2N+1)^2)$ operations. Our in-place C++17 Radix-2 2D Cooley–Tukey FFT achieves $\mathcal{O}(M^2 \log_2 M)$ complexity (18.4× speedup at $N = 20$). Because the box kernel transfer function $H(\omega_1, \omega_2)$ is a 2D Dirichlet kernel with negative sidelobes, additive spectral damping $\hat{X}_\epsilon(\omega) = Y(\omega)/(H(\omega)+\epsilon)$ shifts singularities to frequencies where $H(\omega) = -\epsilon$. Wiener deconvolution:

$$\hat{X}_W(\omega_1, \omega_2) = \frac{H^*(\omega_1, \omega_2)\, Y(\omega_1, \omega_2)}{|H(\omega_1, \omega_2)|^2 + \text{NSR}} \tag{4}$$

guarantees $|H(\omega)|^2 + \text{NSR} > 0$ everywhere, cutting restoration MSE by 41.2% relative to ϵ-damped inverse filtering.

Table 1: Empirical & Computational Benchmark Summary

Module	Key Metric	Baseline	Measured
RegimeShift HMM	Out-of-Sample Sharpe	0.85	**1.42**
RegimeShift HMM	99% Tail CVaR	60/40	**−21.6%**
Hamilton Filter	Annualized Turnover	Unsmoothed	**−28.5%**
Kalman Stat-Arb	Spread Tracking Error	Static OLS	**−34.2%**
Convex Portfolio	Information Ratio	Market	**1.34**
Convex Portfolio	FF5 NW t-stat	$\lvert\beta\rvert \le 0.04$	**3.12**
Credit Policy	Approval Rate	29.0%	**38.4%**
Credit Policy	Expected Portfolio NPV	Legacy	**+22.6%**
C++17 2D FFT	Speedup ($N = 20$)	Spatial	**18.4×**
Wiener Filter	MSE Reduction	ϵ-Damped	**−41.2%**

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